

Curriculum Vitae

Caleb K. Harada

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RESEARCH INTERESTS

Characterization of exoplanets and host stars; planetary habitability; formation and evolution of planetary systems; extrasolar moons and rings; exoplanet atmospheres

EDUCATION

- Ph.D. in Astrophysics**, University of California, Berkeley May 2026
Dissertation: “Exomoons to ExoEarths: Pathways Toward Habitable Worlds Around Sun-like Stars”
Committee: Courtney Dressing (chair), Eugene Chiang, Jennifer Bergner
- M.A. in Astrophysics**, University of California, Berkeley May 2022
- B.S. in Astronomy** (High Honors), University of Maryland, College Park May 2020
Honors Thesis: “Signatures of Clouds in Hot Jupiter Atmospheres”
Advisor: Eliza M.-R. Kempton
Magna Cum Laude
- B.S. in Physics**, University of Maryland, College Park May 2020
Magna Cum Laude

RESEARCH APPOINTMENTS

- **Postdoctoral Researcher**, Institute for Astronomy, University of Hawai'i at Mānoa 2026–Present
- **Graduate Student Researcher**, Dept. of Astronomy, UC Berkeley 2020–2026
- **NSF Graduate Research Fellow**, Dept. of Astronomy, UC Berkeley 2020–2025
- **Undergraduate Student Researcher**, Dept. of Astronomy, University of Maryland 2017–2020
- **NSF REU Intern**, Center for Astrophysics | Harvard & Smithsonian 2019
- **Summer Research Assistant**, Dept. of Astronomy, University of Michigan 2018
- **Summer Research Assistant**, Dept. of Astronomy & Astrophysics, University of Chicago 2017

AWARDS, FELLOWSHIPS & CERTIFICATES

- UC Berkeley Astronomy Diversity, Equity, Inclusion, and Climate (DEIC) Prize 2026
- AAS Rodger Doxsey Dissertation Travel Prize (\$400) 2026
- UC Berkeley Certificate in Teaching & Learning in Higher Education 2025
- UC Berkeley Outstanding Graduate Student Instructor Award (\$1k) 2022
- NSF Graduate Research Fellowship (\$34k–\$37k annually for 3 years) 2020
- UMD University Honors Certificate 2019
- NSF Research Experience for Undergraduates (\$5k) 2019
- Maryland Space Grant Scholarship (\$2k) 2019
- UMD Monroe Martin Undergraduate Research Showcase Award (\$250) 2018, 2019
- UMD Angelo Bardasis Memorial Scholarship (\$2.5k) 2018, 2019
- UMD President's Scholarship (\$5k annually for 4 years) 2016

First Author

21. **Harada, C. K.**, Dressing, C. D., Turtelboom, E. V., *et al.* (2025). “SPORES-HWO. II. Companion Mass Limits and Updated Planet Properties for 120 Future Exoplanet Imaging Targets from 35 yr of Precise Doppler Monitoring.” [AJ](#), **170**, 343. arXiv:2409.10679. DOI:10.3847/1538-3881/ae0b62.
20. **Harada, C. K.**, Dressing, C. D., Kane, S. R., & Ardestani, B. A. (2024). “Setting the Stage for the Search for Life with the Habitable Worlds Observatory: Properties of 164 Promising Planet-survey Targets.” [ApJS](#), **272**, 30. arXiv:2401.03047. DOI:10.3847/1538-4365/ad3e81.
19. **Harada, C. K.**, Dressing, C. D., Alam, M. K., *et al.* (2023). “Stability and Detectability of Exomoons Orbiting HIP 41378 f, a Temperate Jovian Planet with an Anomalously Low Apparent Density.” [AJ](#), **166**, 208. arXiv:2303.14294. DOI:10.3847/1538-3881/ad011c.
18. **Harada, C. K.**, Kempton, E. M.-R., Rauscher, E., *et al.* (2021). “Signatures of Clouds in Hot Jupiter Atmospheres: Modeled High-resolution Emission Spectra from 3D General Circulation Models.” [ApJ](#), **909**, 85. arXiv:1912.02268. DOI:10.3847/1538-4357/abdc22.

Second or Third Author

17. Tuchow, N. W., **Harada, C. K.**, Mamajek, E. E., *et al.* (2025). “HWO Target Stars and Systems: A Prioritized Community List of Potential Stellar Targets for the Habitable Worlds Observatory’s ExoEarth Survey.” [PASP](#), **137**, 104402. arXiv:2509.20544. DOI:10.1088/1538-3873/ae0a81.
16. Desai, A., Turtelboom, E. V., **Harada, C. K.**, *et al.* (2024). “The TESS-Keck Survey. XVIII. A Sub-Neptune and Spurious Long-period Signal in the TOI-1751 System.” [AJ](#), **167**, 194. arXiv:2402.07110. DOI:10.3847/1538-3881/ad29ee.
15. Duck, A., **Harada, C. K.**, Harrell, J., *et al.* (2021). “K2, Spitzer, and TESS Transits of Four Sub-Neptune Exoplanets.” [AJ](#), **162**, 136. arXiv:2108.08889. DOI:10.3847/1538-3881/ac0e2f.

Coauthor

14. Kane, S. R., Bott, K. M., Goodis Gordon, K. E., *et al.*, including **Harada, C. K.** (2026). “Imaging Venus-like Worlds: Spectral, Polarimetric, and UV Diagnostics for the Habitable Worlds Observatory.” [PASP](#), **138**, 024404. arXiv:2602.02728. DOI:10.1088/1538-3873/ae417d.
13. Blunt, S., Nielsen, E. L., Newton, E. R., *et al.*, including **Harada, C. K.** (2025). “Statistical method for constraining the capability of the Habitable Worlds Observatory to understand ozone onset time in Earth analogs.” [JATIS](#), **11**, 042214. arXiv:2507.06188. DOI:10.1117/1.JATIS.11.4.042214.
12. Sagynbayeva, S., Abbas, A., Kane, S. R., *et al.*, including **Harada, C. K.** (2025). “Requirements for Joint Orbital Characterization of Cold Giants and Habitable Worlds with Habitable Worlds Observatory.” [AJ](#), **170**, 208. arXiv:2507.21443. DOI:10.3847/1538-3881/adf84d.
11. Li, Z., Kane, S. R., Blunt, S., & **Harada, C. K.** (2025). “Radial Velocity Strategies for the Orbital Refinement of Exoplanet Direct Imaging Targets.” [PASP](#), **137**, 104401. arXiv:2509.17169. DOI:10.1088/1538-3873/ae0520.
10. Grouffal, S., Santerne, A., Bourrier, V., *et al.*, including **Harada, C. K.** (2025). “The star HIP 41378 potentially misaligned with its cohort of long-period planets.” [A&A](#), **701**, A173. arXiv:2507.01807. DOI:10.1051/0004-6361/202555487.
9. Mayo, A. W., Fortenbach, C. D., Louie, D. R., *et al.*, including **Harada, C. K.** (2025). “Detection of H₂O and CO₂ in the Atmosphere of the Hot Super-Neptune WASP-166b with JWST.” [AJ](#), **170**, 50. arXiv:2501.00609. DOI:10.3847/1538-3881/adda2e.
8. Turtelboom, E. V., Dietrich, J., Dressing, C. D., & **Harada, C. K.** (2025). “Searching for Additional Planets in TESS Multiplanet Systems: Testing Empirical Models Based on Kepler Data.” [AJ](#), **170**, 3. arXiv:2409.03852. DOI:10.3847/1538-3881/adcd71.
7. Wittenmyer, R. A., Errico, A., Holt, T. R., *et al.*, including **Harada, C. K.** (2025). “Optimizing radial velocity detection limits for Southern Habitable Worlds Observatory targets.” [MNRAS](#), **539**, 457. arXiv:2503.20121. DOI:10.1093/mnras/staf503.

6. Kane, S. R., Li, Z., Turnbull, M. C., Dressing, C. D., & **Harada, C. K.** (2024). “Dynamical Viability Assessment for Habitable Worlds Observatory Targets.” *AJ*, **168**, 195. arXiv:2408.00263. DOI:10.3847/1538-3881/ad6a50.
5. Turtelboom, E. V., Weiss, L. M., Dressing, C. D., *et al.*, including **Harada, C. K.** (2022). “The TESS-Keck Survey. XI. Mass Measurements for Four Transiting Sub-Neptunes Orbiting K Dwarf TOI-1246.” *AJ*, **163**, 293. arXiv:2204.11895. DOI:10.3847/1538-3881/ac69e5.
4. Malsky, I., Rauscher, E., Kempton, E. M.-R., *et al.*, including **Harada, C. K.** (2021). “Modeling the High-resolution Emission Spectra of Clear and Cloudy Nontransiting Hot Jupiters.” *ApJ*, **923**, 62. arXiv:2110.05593. DOI:10.3847/1538-4357/ac2a2a.
3. Roman, M. T., Kempton, E. M.-R., Rauscher, E., *et al.*, including **Harada, C. K.** (2021). “Clouds in Three-dimensional Models of Hot Jupiters over a Wide Range of Temperatures. I. Thermal Structures and Broadband Phase-curve Predictions.” *ApJ*, **908**, 101. arXiv:2010.06936. DOI:10.3847/1538-4357/abd549.
2. Vissapragada, S., Knutson, H. A., Jovanovic, N., *et al.*, including **Harada, C. K.** (2020). “Constraints on Metastable Helium in the Atmospheres of WASP-69b and WASP-52b with Ultranarrowband Photometry.” *AJ*, **159**, 278. arXiv:2004.13728. DOI:10.3847/1538-3881/ab8e34.
1. Collins, K. A., Collins, K. I., Pepper, J., *et al.*, including **Harada, C. K.** (2018). “The KELT Follow-up Network and Transit False-positive Catalog: Pre-vetted False Positives for TESS.” *AJ*, **156**, 234. arXiv:1803.01869. DOI:10.3847/1538-3881/aae582.

OTHER MANUSCRIPTS

5. **Harada, C. K.**, Kane, S. R., Dressing, C. D., *et al.* (2026). “RV Detection of an Eccentric Companion to a Nearby Sun-like Star.” In preparation, to be submitted to AAS Journals.
4. Abbas, A., Nielsen, E., Macintosh, B., *et al.*, including **Harada, C. K.** (2026). “Orbit Recovery and Demographics from a Simulated Exo-Earth Survey with the Habitable Worlds Observatory.” Submitted to AAS Journals.
3. Dressing, C. D., Savel, A., Giacalone, S., *et al.*, including **Harada, C. K.** (2026). “Cleaning up the TOIs I: ShARCS Follow-up Adaptive Optics Imaging of 603 TESS Targets.” Submitted to AAS Journals.
2. Flores, B. E., White, R. J., Schaefer, G., *et al.*, including **Harada, C. K.** (2026). “Precise Radii of Visual Binaries with Dynamical Masses: ξ Bootis & 70 Ophiuchi.” Submitted to AAS Journals.
1. Li, Y., Meyer, M. R., D’Angiolillo, S., *et al.*, including **Harada, C. K.** (2026). “A Century of Radial Velocity and Astrometric Monitoring of 70 Oph AB: New PFS Data and Constraints on Planetary Companions.” Submitted to AAS Journals, arXiv:2603.20044. DOI:10.48550/arXiv.2603.20044.

RESEARCH PRESENTATIONS

Contributed talk	AAS Meeting #247 , Phoenix, AZ, USA	Jan. 5, 2026
Invited talk	UC Berkeley CIPS Seminar , Berkeley, CA, USA	Dec. 3, 2025
Invited talk	UC Riverside Astrobiology Seminar , Riverside, CA, USA	Nov. 14, 2025
Invited talk	Caltech/IPAC Lunch Seminar , Pasadena, CA, USA	Nov. 12, 2025
Invited talk	Harvard/CfA Special Seminar , Cambridge, MA, USA	Oct. 16, 2025
Invited talk	UC Santa Cruz PLUNCH Seminar , Santa Cruz, CA, USA	Sep. 29, 2025
Contributed talk	HWO25 Conference , Washington, DC, USA	Jul. 30, 2025
Poster	Sagan Summer Workshop 25 , Pasadena, CA, USA	Jul. 21–25, 2025
Contributed talk	Bay Area Exoplanet Meeting #48 , Santa Cruz, CA, USA	Jul. 18, 2025
Invited talk	HWO TSS-SWG Multiplicity Group Meeting , Remote	Jul. 16, 2025
Poster	EPRV6 Conference , Porto, Portugal	Jul. 1–3, 2025
Poster	Know Thy Star; Know Thy Planet 2 , Pasadena, CA, USA	Feb. 3–7, 2025
Contributed talk	AAS Meeting #245 , National Harbor, MD, USA	Jan. 15, 2025
Invited talk	Yale Exoplanets & Stars Seminar , New Haven, CT, USA	Oct. 30, 2024
Invited talk	MIT Kavli Monday Afternoon Talk , Cambridge, MA, USA	Oct. 28, 2024
Poster	Sagan Summer Workshop 24 , Pasadena, CA, USA	Jul. 22–26, 2024
Poster	Exoplanets 5 Conference , Leiden, Nederland	Jun. 20, 2024

Poster	Extreme Solar Systems V , Ōtautahi, Aotearoa	Mar. 17–21, 2024
Contributed talk	AAS Meeting #243 , New Orleans, LA, USA	Jan. 10, 2024
Invited talk	AAS #243 HWO Splinter Session , New Orleans, LA, USA	Jan. 10, 2024
Invited talk	UC Santa Cruz PLUNCH Seminar , Santa Cruz, CA, USA	Oct. 31, 2022
Contributed talk	UC Berkeley Astronomy Lunch Talk , Berkeley, CA, USA	Oct. 20, 2022
Contributed talk	Bay Area Exoplanet Meeting #36 , Remote (COVID-19)	Mar. 5, 2021
Poster	AAS Meeting #235 , Honolulu, HI, USA	Jan. 6, 2020
Contributed talk	CfA/SAO Summer REU Symposium , Cambridge, MA, USA	Aug. 8, 2019
Poster	Chesapeake Bay Exoplanet Meeting #4 , College Park, MD, USA	Feb. 15, 2019
Poster	AAS Meeting #233 , Seattle, WA, USA	Jan. 8, 2019

APPROVED OBSERVING PROPOSALS

2.4-m Automated Planet Finder (APF), Lick Observatory

- Levy Spectrograph: “Probing Planetary System Architectures Around Neglected Nearby Stars in Preparation for HWO” (PI: **Harada**); 9 nights in 2025A/2025B/2026A
- Levy Spectrograph: “The APF Survey of the Habitable Worlds Observatory’s Neglected Targets” (PI: Holden; Co-I: **Harada**); 8 nights in 2025A/2025B

3-m Shane Telescope, Lick Observatory

- ShARCS AO: “Looking for Close Stellar Companions to Potential Targets for Future Searches for Life with HWO” (PI: Dressing; Co-I: **Harada**); 10 nights in 2025A/2025B
- ShARCS AO: “A Closer Look at the Host Stars of Transiting Planets” (PI: Dressing; Co-I: **Harada**); 25 nights in 2021A/2021B/2022A

10-m Keck I Telescope, Mauna Kea Observatory

- KPF Spectrograph: “Piecing Together the Puzzling TOI-880 System” (PI: Turtelboom; Co-I: **Harada**); 1 night in 2025B
- KPF Spectrograph: “Confirming an Eccentric Sub-Saturn Planet Around the Bright Nearby Star HD 219623” (PI: **Harada**); 0.25 nights in 2025A
- HIRES: “The Obliquity of the HIP 41378 Planetary System” (PI: Dressing; Co-I: **Harada**); 18 hr in 2022B

8.1-m Keck I Telescope, Mauna Kea Observatory

- MAROON-X Spectrograph: “Characterizing TOI 2104, one of the highest-multiplicity TESS systems” (PI: Turtelboom; Co-I: **Harada**); 40 hr in 2024B/2025B

3.5-m WIYN Telescope, Kitt Peak National Observatory

- NEID Spectrograph: “Characterizing TOI 2104, one of the highest-multiplicity TESS systems” (PI: Turtelboom; Co-I: **Harada**); 18 hr in 2024B

1-m Nickel Telescope, Lick Observatory

- Direct Imaging Camera: “Validating Netptune-size planets around A-type stars” (PI: Giacalone; Co-I: **Harada**); 43 nights in 2021B/2022A/2022B

James Webb Space Telescope (JWST)

- NIRISS/NIRSpec: “Transmission Spectroscopy of the Super-Neptune WASP-166b” (PI: Mayo; Co-I: **Harada**); 23 hr in Cycle 1 GO under Program 2062

PROFESSIONAL SERVICE

- **Journal Referee**, A&A; AJ; PASP 2025–Present
- **AAS Chambliss Poster Competition Judge**, AAS Winter Meetings 2024, 2025, 2026
- **HWO Catalogs & Databases Co-Lead**, NASA HWO Target Stars & Systems Group 2024–2025
- **HWO Target Stars & Systems Liaison**, NASA HWO Exoplanet Demographics Group 2024–2025

DEPARTMENTAL SERVICE

- **Co-Organizer**, UC Berkeley AstroQ (community for LGBTQ+ astronomers) 2023–2026
- **Grad Representative**, UC Berkeley Astrophysics Graduate Admissions Committee 2025, 2026
- **Grad Representative**, UC Berkeley Astronomy DEI & Climate Committee 2021–2022
- **Grad Representative**, UC Berkeley Astro Grad Prospective Visit Planning Committee 2021

TEACHING EXPERIENCE

The Planets (UC Berkeley/undergrad non-majors)

- Guest Lecturer Apr. 1, 2025
- Head Graduate Student Instructor Spring 2023

Introduction to Astrophysics (UC Berkeley/undergrad majors)

- Guest Lecturer Nov. 14, 2024, Sep. 5, 2023, Oct. 18, 2022
- Graduate Student Instructor Fall 2021

Atmospheric Physics and Dynamics (UC Berkeley/undergrad majors)

- Course Reader Fall 2022

Introductory Astrophysics II (University of Maryland/undergrad majors)

- Lab Teaching Assistant—remote due to COVID-19 Spring 2020
- Course Tutor Spring 2018

Observational Astronomy (University of Maryland/undergrad majors)

- Lab Teaching Assistant Fall 2018, 2019

MENTORING PROGRAMS

UC Berkeley Astro Grad Mentoring (mentoring program for 1st-year grad students)

- Senior Graduate Student Mentor 2025–2026

SPORES-HWO RV Hunters (community science project for UC Berkeley undergrads)

- Lead Organizer and Mentor Spring 2025

UC Berkeley MPS Scholars (mentoring program for UC Berkeley undergrads)

- Graduate Student Mentor 2024–2025
- Workshop Panelist Dec. 3, 2024, Oct. 10, 2024, Aug. 8, 2024

Cal-Bridge (bridge program for underrepresented undergrads in California)

- Graduate Student Tutor Fall 2024

POWER Bay Area (mentoring program for underrepresented community college students)

- Mentor Coordinator 2022–2023

UMD Astronomy Peer Mentoring (mentoring program for UMD undergrads)

- Undergraduate Student Mentor 2018–2020

OUTREACH PROGRAMS

Letters to a Pre-Scientist (outreach program for middle school students in low-income communities)

- Pen Pal (letter writer) 2024–Present

Astronomy on Tap (public lecture series)

- Guest Speaker, Two Pitches Brewing Company, Oakland, CA Apr. 14, 2026

The Compass Project (UC Berkeley talk series for students in the physical sciences)	
• Guest Lecturer	Oct. 25, 2024
Scientist in Every Florida School (K–12 outreach program for Florida public schools)	
• Guest Speaker, Sunset Park Elementary, Miami, FL	Oct. 7, 2024
University of Maryland Observatory (bimonthly public open house events and observing)	
• Observatory Staff Member	2017–2020
• Observatory Volunteer	2016–2017
UMD Maryland Day (campus-wide UMD public outreach event)	
• Astronomy Department Volunteer	2017–2019

STUDENTS ADVISED

- Christopher Chin, UC Berkeley undergrad (began 2025)
- Aidan Fernandez, USC & Mt. San Antonio Community College undergrad (began 2025)
- Shivani Kulkarni, UC Berkeley undergrad (began 2025)
- Emerald Lin, UC Berkeley undergrad (began 2025)
- Nykole Liu, UC Berkeley undergrad (began 2025)
- Remy Liu, UC Berkeley undergrad (began 2025)
- Abhi Nathan, UC Berkeley undergrad (began 2025)
- Bahareh A. Ardestani, Sonoma State University undergrad (began 2023; co-advised)
- Ryan Hwangbo, UC Berkeley undergrad (began 2023; co-advised)
- Anmol Desai, UC Berkeley undergrad (began 2022; co-advised)

Updated June 2026.